

Logistic Regression

Classification

Classification

Email: Spam / Not Spam?

Online Transactions: Fraudulent (Yes / No)?

Tumor: Malignant / Benign ?

$$y \in \{0, 1\}$$

0: “Negative Class” (e.g., benign tumor)

1: “Positive Class” (e.g., malignant tumor)

Binary Classification



—————→ 1 (cat) vs 0 (non cat)

		Blue			
Green		255	134	93	22
Red	255	134	202	22	2
	255	231	42	22	4
	123	94	83	2	192
	34	44	187	92	34
	34	76	232	124	94
	67	83	194	202	

Classification: $y = 0$ or 1

$h_{\theta}(x)$ can be > 1 or < 0

Logistic Regression: $0 \leq h_{\theta}(x) \leq 1$

Logistic Regression

Hypothesis Representation

(How the function will look like?)

Logistic Regression Model

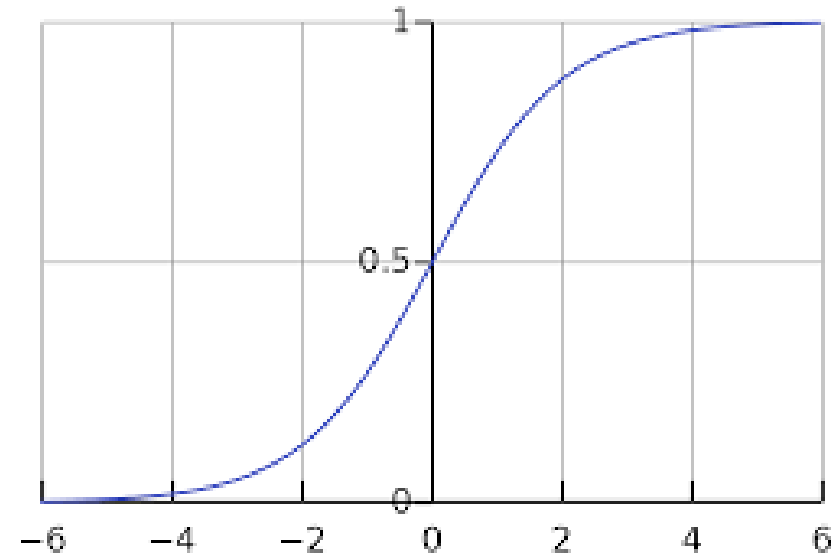
Want $0 \leq h_{\theta}(x) \leq 1$

$$h_{\theta}(x) = \theta^T x$$

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

Sigmoid function
Logistic function



Interpretation of Hypothesis Output

$h_{\theta}(x)$ = estimated probability that $y = 1$ on input x

Example: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$$h_{\theta}(x) = 0.7$$

Tell patient that 70% chance of tumor being malignant

“probability that $y = 1$, given x ,
parameterized by θ ”

$$P(y = 0|x; \theta) + P(y = 1|x; \theta) = 1$$

$$P(y = 0|x; \theta) = 1 - P(y = 1|x; \theta)$$

Logistic Regression

Decision Boundary

(Talking about the decision boundary will give us a better sense of what the logistic regression hypothesis function is computing.)

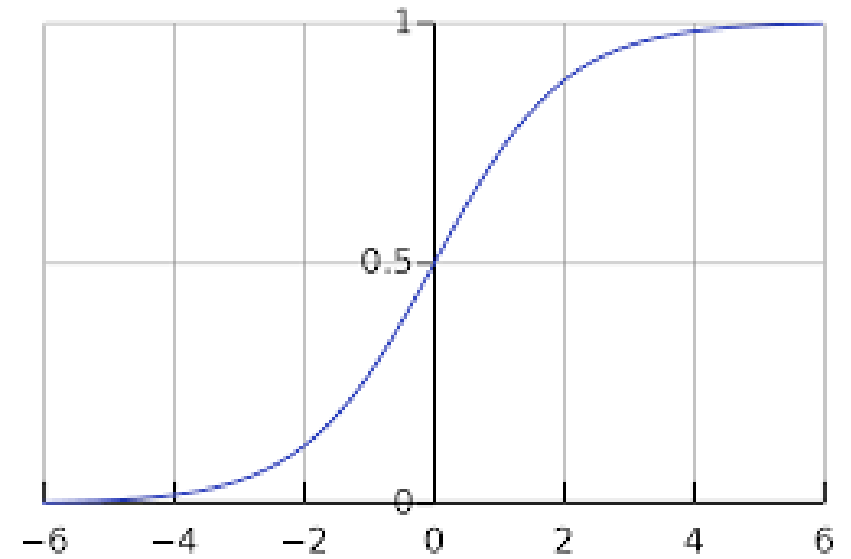
Logistic regression

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1+e^{-z}}$$

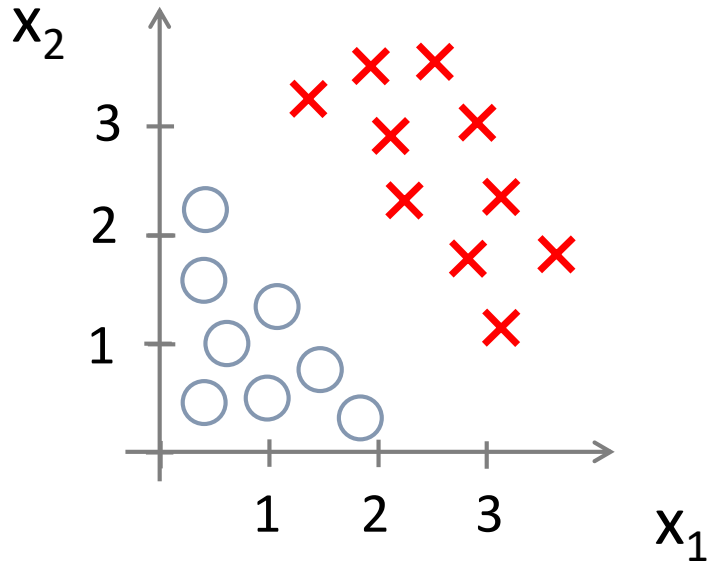
Suppose predict “ $y = 1$ ” if $h_{\theta}(x) \geq 0.5$

predict “ $y = 0$ ” if $h_{\theta}(x) < 0.5$



Let's use this to better understand how the hypothesis of logistic regression makes those predictions.

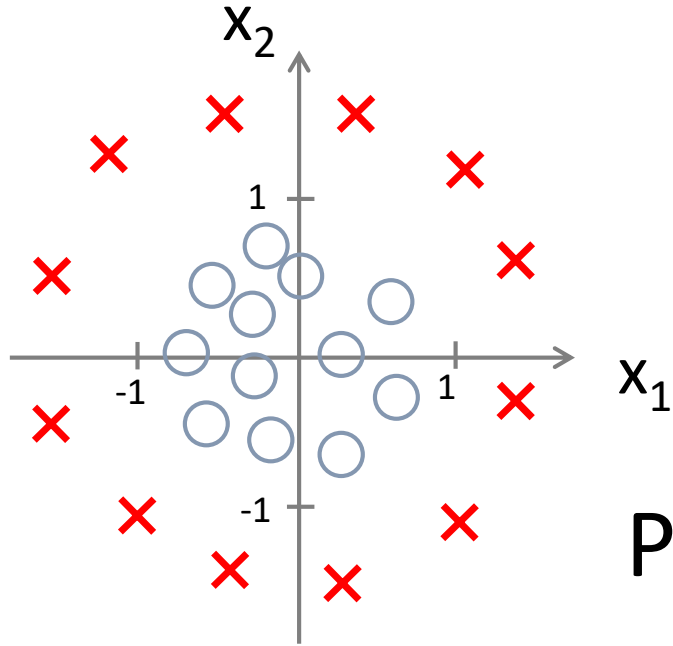
Decision Boundary



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Predict “ $y = 1$ ” if $-3 + x_1 + x_2 \geq 0$

Non-linear decision boundaries



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

Predict “ $y = 1$ ” if $-1 + x_1^2 + x_2^2 \geq 0$

Now that we know what $h(x)$ can represent.

Next: see how to automatically choose the parameters θ .

So that given a training set we can automatically fit the parameters to our data.

Logistic Regression

Cost Function

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

m examples $x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix} \quad x_0 = 1, y \in \{0, 1\}$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

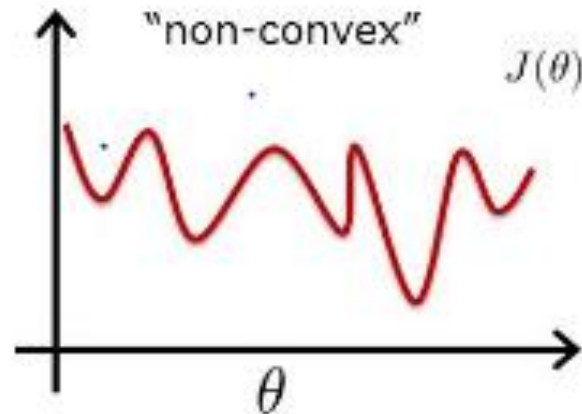
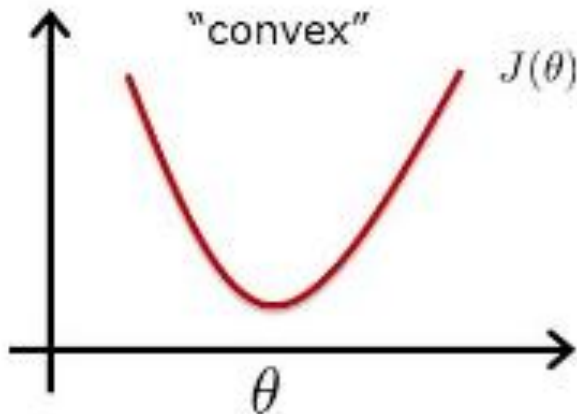
How to choose parameters θ ?

Cost function

Linear regression: $J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$

$$\text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) = \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

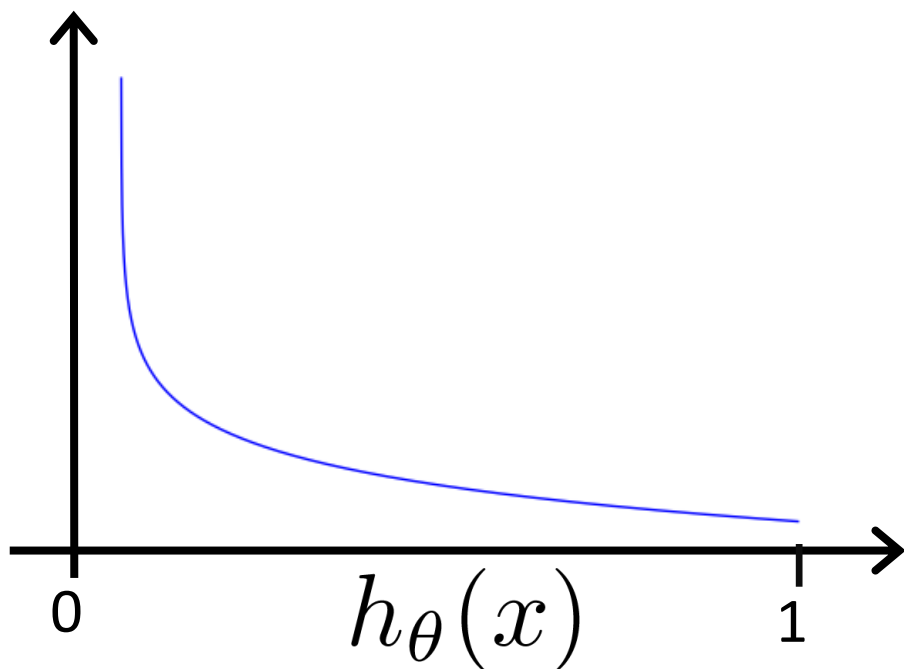
Convex Vs Non-Convex



Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

If $y = 1$



Cost = 0 if $y = 1, h_{\theta}(x) = 1$

But as $h_{\theta}(x) \rightarrow 0$

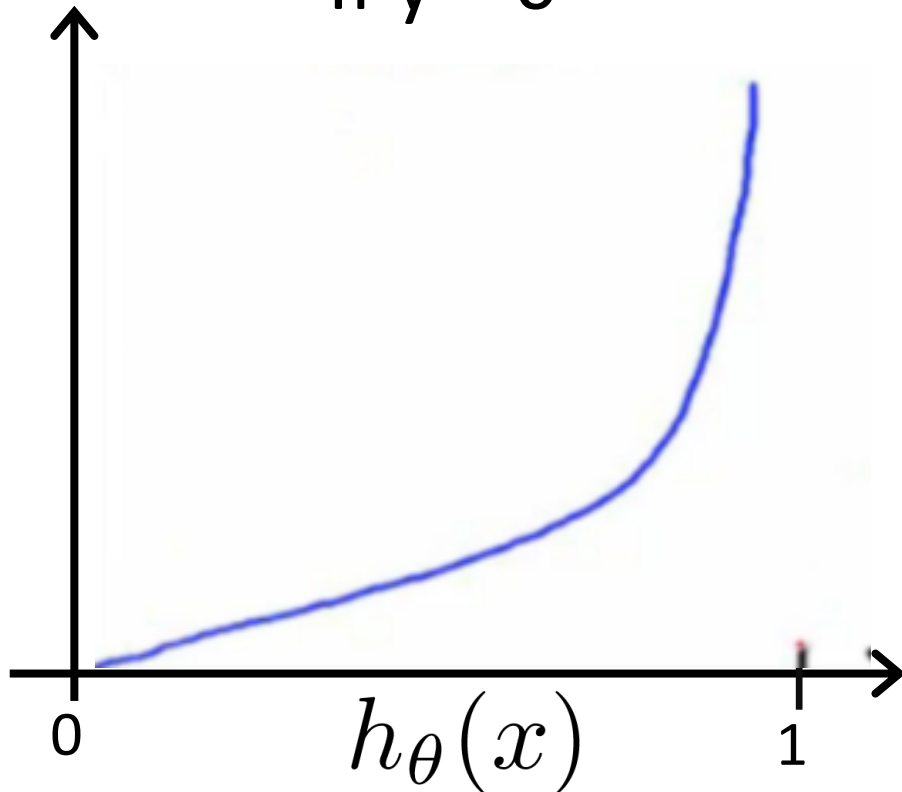
$\text{Cost} \rightarrow \infty$

Captures intuition that if $h_{\theta}(x) = 0$, (predict $P(y = 1|x; \theta) = 0$), but $y = 1$, we'll penalize learning algorithm by a very large cost.

Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

If $y = 0$



Logistic Regression

Cost Function and Gradient Descent

Logistic regression cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Note: $y = 0$ or 1 always

Logistic regression cost function

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] \end{aligned}$$

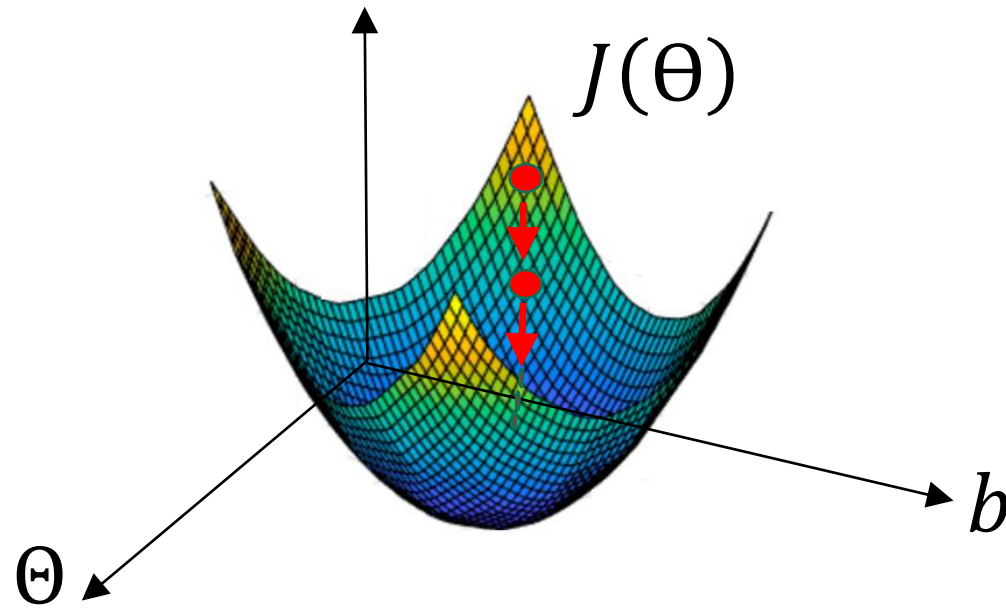
To fit parameters θ :

$$\min_{\theta} J(\theta)$$

To make a prediction given new x :

$$\text{Output } h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Gradient Descent



Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

(simultaneously update all θ_j)

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

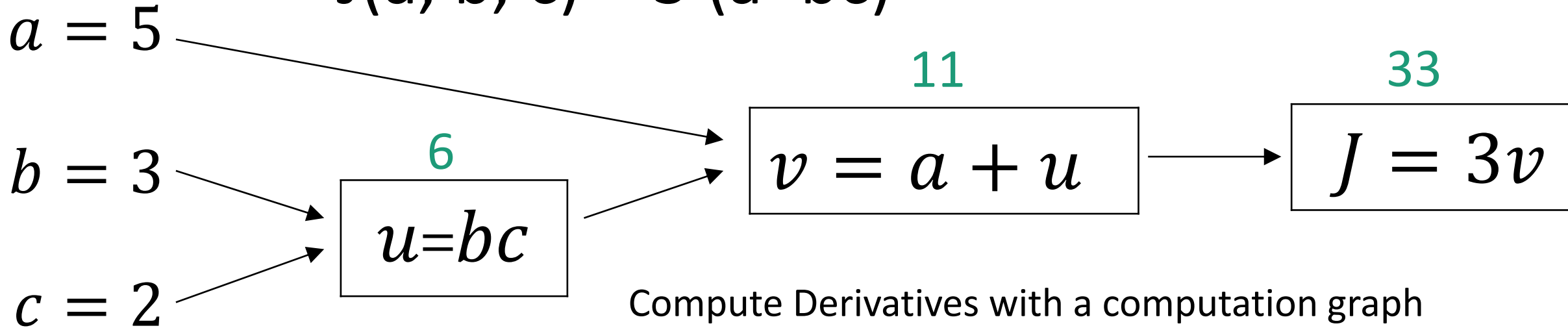
$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

} (simultaneously update all θ_j)

Algorithm looks identical to linear regression!

Computation Graph

$$J(a, b, c) = 3(a + bc)$$



$$\frac{\partial J}{\partial u} = \frac{\partial J}{\partial v} * \frac{\partial v}{\partial u}$$

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial u} * \frac{\partial u}{\partial b}$$

$$\frac{\partial J}{\partial c} = \frac{\partial J}{\partial u} * \frac{\partial u}{\partial c}$$

$$\frac{\partial J}{\partial a} = \frac{\partial J}{\partial v} * \frac{\partial v}{\partial a}$$

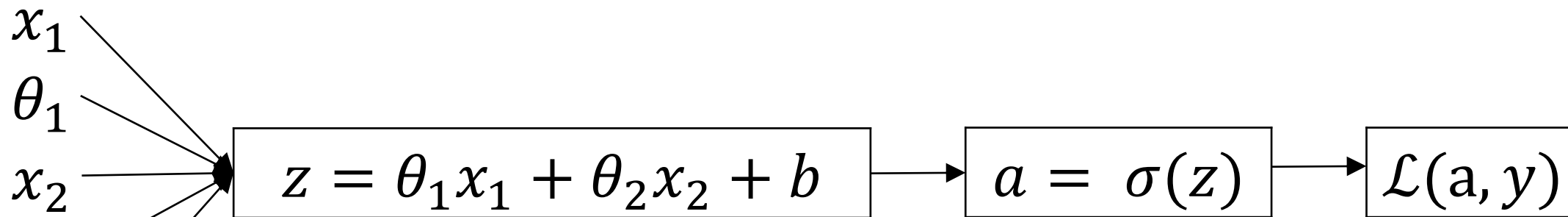
Computation Graph: Logistic Regression and Gradient

$$z = \theta^T x + b$$

$$\hat{y} = a = \sigma(z)$$

$$\mathcal{L}(a, y) = -(y \log(a) + (1 - y) \log(1 - a))$$

Logistic regression derivatives



$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial a} * \frac{\partial a}{\partial z} = a - y$$

$$\frac{\partial a}{\partial z} = a(1 - a)$$

$$\frac{\partial \mathcal{L}}{\partial a} = -\frac{y}{a} + \frac{1 - y}{1 - a}$$

Logistic regression on m examples

$$J(\theta, b) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(a^i, y^i)$$

$$a^i = \hat{y} = \sigma(z^i) = \sigma(\theta^T x^i + b)$$

$$\frac{\partial}{\partial \theta_i} J(\theta, b) = \frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial \theta_i} \mathcal{L}(a^i, y^i)$$

Logistic regression on m examples

$J=0$; $d\Theta_1=0$; $d\Theta_2=0$; $db=0$

For $i=1$ to m

$$z^i = \Theta^T x^i + b$$

$$a^i = \sigma(z^i)$$

$$J += -[y^i \log(a^i) + (1 - y^i) \log(1 - a^i)]$$

$$dz^i = a^i - y^i$$

$$d\Theta_1 += x_1^i * dz^i$$

$$d\Theta_2 += x_2^i * dz^i$$

$$db += dz^i$$

$$J = J/m$$

$$d\Theta_1 /= m$$

$$d\Theta_2 /= m$$

$$db /= m$$